

REPUBLIQUE DU CAMEROUN
PAIX – TRAVAIL – PATRIE

REPUBLIC OF CAMEROON
PEACE – WORK – FATHERLAND

MINISTERE DE L'ENSEIGNEMENT SUPERIEUR
INSTITUT UNIVERSITAIRE DE LA COTE

MINISTRY OF HIGHER EDUCATION
UNIVERSITY INSTITUTE OF THE COAST



RESEARCH PROJECT

DESIGN OF AN INDUSTRIAL SHREDDER PAPER MACHINE

From the 28th of June to the 27th of August

DEPARTMENT: BACHELOR OF TECHNOLOGY

SPECIALTY: MECHANICAL MANUFACTURING

LEVEL: LEVEL 300

WRITTEN BY: TCHEMO ISRAEL JUNIOR

MATRICULE NUMBER: FPSE2267

Under the supervision of:

Academic supervisor:

Mr. BESONG PLATINI

ACADEMIC YEAR 2023/2024

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Abstract:

A paper shredder is considered to be a mechanical machine that is used to cut paper into either strips or fine particles that portray no information that was written on the paper initially. Government establishments, big business holders, and private personalities use these shredders to abolish private, personal, or otherwise delicate and secret pamphlets. Privacy specialists frequently claim that individuals should shred their bills, tax papers, credit cards, and statements of their bank account, and other kinds of stuff that might be used by robs to commit deceit and theft. In this report, we describe the design and working of a shredder machine that would be able to shred industrial paper and soft solid.

LIST OF TABLES

Table 1. SPECIFICATION.....	11
Table 2.ENGINEERING DESIGN STANDARD.....	19
Table 3. SUMMARY OF RESULT.....	30
Table 4.COST ESTIMATION.....	47
Table 5.SOUND & VIBRATION RESULT	48
Table 6.DECISION MATIX ANALYSIS.....	49
Table 7.COST ANALYSIS	49
Table 8. MAINTENANCE LOG	61

LIST OF FIGURES

Figure 1.sketch of shredder machine	12
Figure 2.shredder machine	12
Figure 3.transmission system.....	33
Figure 4.isometric view of shredder machine	33
Figure 6.view point of a shredder.....	35
Figure 7.assembly	35
Figure 8.cutting system	39
Figure 9.pulley drive.....	40
Figure 10.gear drive	40
Figure 11.bearing.....	40
Figure 12.shaft	41
Figure 13.pulley	41
Figure 14.material list	44
Figure 15.rigth side view & front view	45

Table of Contents

Abstract:	1
Acknowledgments	5
Chapter 1: Introduction.....	6
1.1 Project Definition	6
1.2 Project Objectives	7
1.3 Project Specifications	8
1.4 Applications	10
Chapter 2: Literature Review.....	11
2.1 Project background.....	11
2.2 Previous Work.....	11
2.3 Comparative Study.....	13
Chapter 3: System Design	14
3.1 Design Constraints and Design Methodology	14
3.1.1 Geometrical Constraints	14
3.1.2 Sustainability	14
3.1.3 Environmental	14
3.1.4 Social	15
3.1.5 Economic	15
3.1.6 Safety	15
3.1.7 Ethical.....	15
3.2 Engineering Design standards	16
3.2.1 Shaft.....	16
3.2.2 Cutting Blades	17

3.2.3 Screws.....	17
3.2.4 Motor	18
3.2.5 Gears.....	18
3.3 Theory and Theoretical Calculations	21
3.3.1 Gears Rotational Speed.....	21
3.3.2 Design of Shaft	21
3.3.3 Torque on Shaft	21
3.3.4 Cutting Force of Industrial Paper.....	22
3.4.5 Number of Papers in Each Cut	22
3.4.6 Results	22
3.4.7 Summary of the Calculations.....	24
3.4 Product Subsystems and selection of Components.....	24
3.5) Manufacturing assembly study — — — — —	31
3.6 3D CAD DESIGN	28
3.7 Economic Evaluation of projects.....	28
Chapter 4: System Testing & analysis	32
4.1 Sounds and vibration test results [9][10]	32
4.2 Decision matrix [10][11]	32
4.3 Cost analysis	33
Chapter 5: Project Management	35
5.1 Project planning & Monitoring.....	35
5.2 Team members contribution	35
5.3 Project Challenges	0
5.3.1 Design challenges overview	0
5.3.2 Fabrication challenges overview.....	0
Chapter 6: Project Analysis	1

6.1 Long term learning	1
6.2 Impact of Engineering Solutions	1
6.3 Contemporary Issues Addressed.....	2
Chapter 7: Conclusions and Future Recommendations	3
7.1 Conclusions	3
7.2 Future Recommendations.....	3
7.3) MAINTENANCE PLAN -----	59

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Chapter 1: Introduction

The shredding machine was designed to solve problems on identity theft, fraud, cost of disposing of paper, and ease of paper recycling by properly shredding sensitive documents. This report includes a complete study of different parts of the shredder machine such as the frame of the shredder, the system of transmission, and the cutting system that are made and designed independently. The first part of the report incorporates the study of the cutting system of the shredder machine for example types of blades, advantages and drawbacks of different kinds of blades. The second part includes the process of design with the comprehensive design of each element of the cutting and transmission system i.e., blade designing and making certain alterations in it and the frame of the machine.

1.1 Project Definition

This report explains a detailed study and designing method of a shredder machine that would be used to shred the industrial paper and soft solid. The report comprises a complete study of various parts of the shredder machine for example the frame of the shredder, the system of transmission, and the cutting system that are made and designed independently.

Categories of paper cutting shredders:

Strip-cut shredders:

In the strip-cut shredder machine, the revolving blades were used to cut thin strips as long as the unique sheet of paper. These types of strips could be reunited.

Cross-cut shredders: -

In these types of shredders, there are two opposite revolving drums to cut four-sided, rectangle, or diamond-shaped shreds.

Particle-cut shredders:

These types of shredder produce minute square or circular fragments.

Cardboard type shredders:

These types are planned particularly to shred grooved material into either strips or a network of the pallet.

Pierce-and-tear type shredders:

These shredders are composed of spinning knife-edges that penetrate the paper and then tear it away from each other.

Grinder type shredders:

They are composed of a rotating shaft with cutting knife-edges that chore the paper until it is small enough to drop from a screen.

Disintegrators and granulator type shredders:

These are used to constantly cut the paper haphazard until the constituent parts are minor enough to pass through a porous plate.

1.2 Project Objectives

To study the diverse parts of the shredder machine for example the knife-edges, the frame, the transmission system, etc.

To plan a shredding machine that will yield less sound and vibrations.

To design a machine that will shred industrial paper and soft solid.

To keep the cost of structure as little as possible with no compromise on the final yield.

1.3 Project Specifications

Specifications:

The specifications of the machine are:

Table 1. SPECIFICATION

Vertical Shredder	VS100	VS300	VS600
Motor:	Siemens	Siemens	Siemens
Power:	70 Kw	120 Kw	330 Kw
Voltage:	As per request	As per request	As per request
	2500*3000*4200 mm	3800*3200*4800 mm	4400*3000*5800 mm
Chamber Height	600 mm	800 mm	900 mm
Rotor Diameter	φ1100 mm	φ1150 mm	φ1150 mm
Equipment Weight	2000 Kg	1500 Kg	1500 Kg
Final Product	20-100mm	30-100mm	30-100mm
	1200-1500 Kg/Hr	2000-3000 Kg/Hr	3000-4000 Kg/Hr

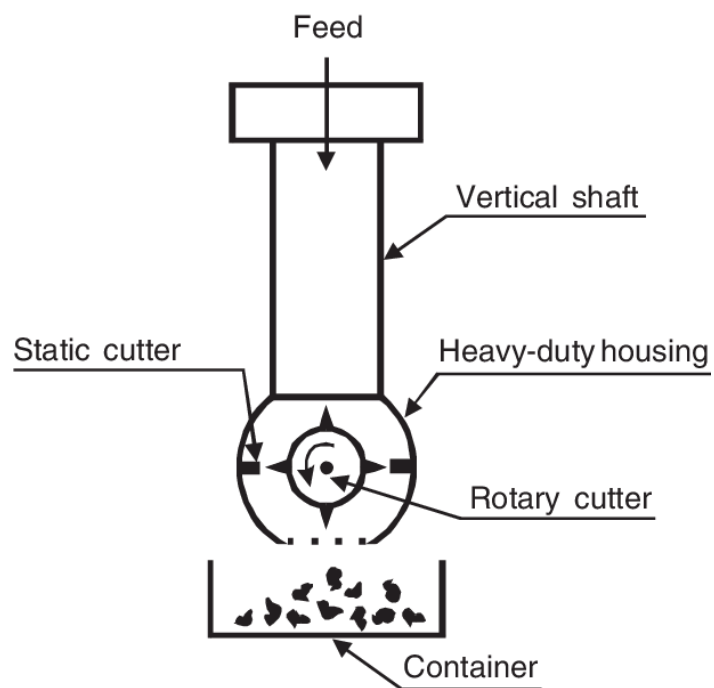
Diagram:

Figure 1.sketch of shredder machine

Real-life picture:

Figure 2.shredder machine

1.4 Applications

Following are the main applications of the shredder we are going to design:

To terminate the information on the business or important organizational documents.

To make the fine chips of the solid waste.

To shred the industrial paper.

The industrial paper after shredding may be used by recycling process

Chapter 2: Literature Review

2.1 Project background

This paper includes a complete study of different parts of the shredder machine such as the frame of the shredder, the system of transmission, and the cutting system that are made and designed independently. The first part of the report incorporates the study of a cutting system of the shredder machine for example types of blades, advantages and drawbacks of different kinds of blades. The second part includes a process of design with the comprehensive design of each element of the cutting and transmission system i.e., blade designing and making certain alterations in it and the frame of the machine.

2.2 Previous Work

In 2000 Joseph Y. Ko presented a shredder machine with instinctive feeding apparatus which was capable to shred twenty sheets with approximately nine inches width. This machine had three ways to switch for example On, Off, and Auto mode. The blades of the machine were knife rollers which cut the strips of paper but can be seldom organized to have confetti-cuts of the paper need to be shredded. The feeding apparatus had a pair of rollers to guide the paper. The rollers and the knife blades of the shredding machine were driven by a single alternating current motor and a belt and pulley mechanism. In 2000 Frank Chang presented the assembly of blades for a paper shredder that was in a contrasted way. Conventional assembly was composed of extended and short rings. The drawback of the shredder was that even if one part failed, the entire assembly gets released. Instead of composing partition rings, it has consisted of elongated and short plates which were cast with the blade ring. The blades were settled on

the rotating shaft to formulate a bladed shaft with the long and short projecting plates of next-to-next blades. This preparation eradicated the use of partition rings, plummeting the cost and increasing the assembly efficiency. Gu-Ming Zeng in the year 2006 presented the knife-edges of the paper shredder that had jagged cutting ends which were shaped by twisting. This could be completed by utilizing two approaches. In the 1st technique, there was a blade body and notched edge intrinsically shaped and pressed from the same fundamental material. With the use of this apparatus, the cost of manufacture was high and even high-level solid material was mandatory for the operation of the machine. On the other hand, in the second technique, there were notched cutting ends particularly condensed to lessen material eating. The problem was that they were also difficult to produce. Three different kinds of blades could be considered according to his approach in the first kind 4 serrated edges regularly spaced with one annular overhang. The second kind of serrated ends with corrugations on the external margin. In the third kind 2 serrated ends with three annular flanges squarely spaced. Ming- Hui Ho. in the year 2003 gave the idea of the paper shredder which was composed of two rotating cutters each of the shredder with several blades. Every blade of the shredder was made of a first cutting blade with numerous first cutting ends and a second cutting blade with numerous cutting ends. The first and the second cutting edges the both were scattered in a non-equiangular way and each of the first cutting edges was a counterpoise to each one of the second cutting blades in the simple language there was only one cutting edge that was connected with the paper was needed to be shredded. When the quantity of shredded paper enlarged then in this case paper shredder was unable to function ordinarily because several cutting edges instantaneously involved with the paper to be shredded, in this way the paper stuck in the shredder and the functioning of the machine was disturbed. This problematic situation was removed out with the use of a rotating cutter with numerous blades with several cutting ends. With this plan which is described here the sound of the shredding was significantly condensed. Willi

Strohmeyer in the days of 1995 provided us with a blade and a stripper assemblage for a paper shredding machine. In this technique in between the blades of every shaft in the cutter region, stripper bars or fingers were provided to avoid the cutting material that got collected everywhere around the blade of the shaft. In this method, the stripper lump was composed of the row of stripper's fingers that were incorporated in the spaces among the blades of the shredder. Mandatory fidelity was achieved since the fingers were involved with the provision ribs of the contradictory housing. Stripper lump was a booster detail part, thus was modest structure and easy to manufacture, and also had little cost.

2.3 Comparative Study

In this technique in between the blades of every shaft in the cutter region, stripper bars or fingers were provided to avoid the cutting material that got collected everywhere around the blade of the shaft. In this method, the stripper lump was composed of the row of stripper's fingers that were incorporated in the spaces among the blades of the shredder. Mandatory fidelity was achieved since the fingers were involved with the provision ribs of the contradictory housing. Stripper lump was a booster detail part, thus was modest structure and easy to manufacture, and also had little cost.

Chapter 3: System Design

3.1 Design Constraints and Design Methodology

3.1.1 Geometrical Constraints

While designing a soft solid shredder machine, some assumptions are considered that are standard for their parts. The sufficient design constraints were taken into account for efficient, economical, and durable manufacturing. Through proper design calculations and formulation of each part, we made a safe design for this shredder machine. The material selection is based on durability, availability, ease of fabrication, and cost were also considered. When we started the plan and design of our project, we faced numerous challenges such as the unavailability of parts like bearings to assist the rotational motion. In the local mechanical market, we found out low-quality bearings which reduced the efficiency and performance of shredder machines. We designed a shredder machine with minimum weight because we can install it easily in any place.

3.1.2 Sustainability

In this shredder machine project, we faced a sustainability problem due to the possibility of deformation and corrosion. Few important steps must be taken to ensure the level of performance and longevity of the machine. These steps include regular repairs and checkups that emptying shredder paper, cleaning, and lubrication to make sure that the parts are running free at their peak performance level.

3.1.3 Environmental

We need to reduce the amount of unrecyclable waste to promote recycling mechanisms regarding the protection of the environment. The paper shredder is an initial step in the paper

recycling process, in which paper reduces into small pieces before reaching reprocessing centers. With the help of a paper shredder process, the reprocessing done too easier.

3.1.4 Social

Most private companies, government organizations, and individual persons care about their confidential and private information, which must be present in paper format. The shredder machine destroying secret information and preventing it from falling into unwanted hands. However, such type of system has a social impact because provided a sense of safety and security.

3.1.5 Economic

There numerous factors as machine cost, labor cost, fuel, and transport cost are mainly depending on the paper production cost. The demand for new industrial paper will be reduced, with the help of the recycling and shredding process because production cost automatically decreases correspondingly.

3.1.6 Safety

Privacy and safety information is important reasons for using a paper machine. Shredding of private or confidential documents ensures that there will be no traces left for unwanted eyes to take look at.

3.1.7 Ethical

Now a day, companies, businesses, peoples, and societies with their secret information, with confidence that their data is going to be kept safe. However, we knowns that the business shredder is an important technique of discarding documents without bargaining client's security. The paper shredder technique promises to their customers by confirming that their secret information would not have seeped.

3.2 Engineering Design standards

In this shredder project, each component is followed the engineering design standard. In this heading, we enlisted that each component has been selected for our shredder project. The shredder machine frame compromises feed unit, machine stand, cutting unit, and hopper housing respectively. The selected component includes a shaft, cutting blades, gears, screws, motor, and fans. The bearing standards have been taken according to NSK. The motor standard has been taken according to ISO 9001. The screw standard has been taken according to ANSI Metric.

Table 2. ENGINEERING DESIGN STANDARD

Shredder Components	Engineering Standard	Details
Motor	ISO 9001	U5420
Bearing	NSK	Deep groove ball bearing (6300ZZ)
Blades and Shaft	Cutting standard	DINP-4
Screws	ANSI Metric	B 18.6.3, Fed. Spec. FF-S-92, Fed. Spec. QQ-P-416, MS35206-207, NASM35206

3.2.1 Shaft

In this shredder project, the cylindrical shaft is made with mild steel material. The shaft is

stepped down on both sides with the help of a lathe machine.

Length = 600 mm

Diameter = 35 mm

Stepped diameter = 25 mm

3.2.2 Cutting Blades

In this industrial shredder project, there are 56 blades in each shaft and 112 cutting blades in both shafts respectively. The cutting blades are made with mild steel material. These cutting blades are further divided into two parts as the movable blade and fixed blade. These blades are designed to cut industrial paper with dimensions equal to or less than 160mm^2 . These industrial papers are easier to recycle because we compared their relative size with high- security standards.

3.2.3 Screws

Two types of screws have been used in this shredder design project.

Screw's type 1:

Length: 13.0 mm

Head Height: 1.5 mm

Head Diameter: 4.0 mm

Driver style: Phillips

Screw's type 2:

Length: 9.0 mm

Head Height: 2.0 mm

Head Diameter: 6.0 mm

Driver style: Phillips

3.2.4 Motor

The specifications of the motor are enlisted below:

Motor speed: 6690RPM

Motor power: 0.588kW

Shaft teeth: helical type

Shaft length: 14.0 mm

Diameter: 45.0 mm

Length: 96.0 mm

Armature length: 26.0 mm

Armature width: 54.0 mm

3.2.5 Gears

In industrial paper and soft solid shredder projects, the four types of gear have been used as motor gear, first gear, second gear, and third gear. In this shredder project we used helical gears due to the following reasons:

Helical gears can handle high RPM coming from an electric motor.

Helical gears are used for noiseless power transmission purposes because their teeth are inclined to the axis of the shaft.

Motor Gear:

The motor gear is a double type gear whose side A directly connected to motor gear and side B is connected to gear one respectively.

Side A

Type: Helical Gear

Root diameter: 28 mm

Tooth depth: 2 mm

Thickness: 11 mm

Teeth count 37 teeth.

Tip diameter: 32 mm

Side B

Type: Spur gear.

Tooth depth: 3 mm

Teeth count 7 teeth.

Thickness: :10 mm

Root diameter: 9 mm

Tip diameter: 15 mm

First Gear:

The first gear is a double type gear whose side A connected to side B of the first gear and side B is directly connected to side A of the second gear.

Side A

Type: Spur gear

Tooth depth: 3 mm

Teeth count 28 teeth.

Thickness: 7.5 mm

Root diameter: 38 mm.

Tip diameter: 45 mm

Side B

Type: Spur gear.

Teeth count 6 teeth.

Tooth depth: 3 mm

Thickness: 15 mm

Root diameter: 6.5 mm.

Tip diameter: 13 mm

Second Gear:

The second gear is also a double-type gear that is directly attached to the first shaft.

Side A

Side A is connected to shaft B of the first gear, which was responsible for motion.

Type: Spur gear.

Teeth count 41 teeth.

Tooth depth: 3 mm

Thickness: 4 mm

Root diameter: 57 mm.

Tip diameter: 63 mm

Side B

Side B is directly connected to the third gear.

Type: Spur gear

Tooth depth: 4 mm

Teeth count 9 teeth.

Thickness: 6 mm

Root diameter: 15.5 mm.

Tip diameter: 23.5 mm

Third Gear:

The third gear is responsible for the motion of the second gear shaft, which was directly connected to side B of the second gear.

Type: Spur gear.

Tooth depth: 4 mm

Teeth count 9 teeth.

Thickness: 6 mm

Root diameter: 15.5 mm.

Tip diameter: 23.5 mm

3.3 Theory and Theoretical Calculations

3.3.1 Gears Rotational Speed

The motor shaft is directly connected to the first gear (N_1), and first gear is further connected with the second gear (N_2), and the second gear is again connected to the third gear (N_3). The speed of industrial paper shredder is found out by using a third gear (N_3), which was directly connected to the driven shaft. The industrial paper shredder speed is found out by using the following relation:

$$\text{INDUSTRIAL PAPER SPEED} = \text{MOTOR SPEED} \times \frac{\text{PRODUCT OF TRANSMISSION TEETH}}{\text{PRODUCT OF RECEIVING TEETHS}} \quad (3.1)$$

3.3.2 Shaft PARAMETERS

The material selection for the shaft is AISI 304 standard stainless steel. The standard stainless steel is widely used because of its simplicity and less cost. The torsional moment of the shaft is given as:

$$M_t = \frac{\pi}{16} \tau \quad (3.2)$$

3.3.3 Torque on Shaft

The torque on the shaft is depending on the speed of the industrial paper shredder and motor power respectively. The torque on the shaft is found out by using the following relation:

$$P = \frac{2\pi NT}{16} \quad (3.3)$$

A shaft is a rotating machine element which is used to transmit power from one place to another. The power is delivered to the shaft by some tangential force and the resultant torque (or twisting moment) set up within the shaft permits the power to be transferred to various machines linked up to the shaft. In order to transfer the power from one shaft to another, the various members such as pulleys, gears etc., are mounted on it. These members along with the forces exerted upon them causes the shaft to bending. In other words, we may say that a shaft is used for the transmission of torque and bending moment. The various members are mounted on the shaft by means of keys or splines.

Notes: 1. The shafts are usually cylindrical, but may be square or cross-shaped in section. They are solid in cross-section but sometimes hollow shafts are also used.

2. An axle, though similar in shape to the shaft, is a stationary machine element and is used for the transmission of bending moment only. It simply acts as a support for some rotating body such as hoisting drum, a car wheel or a rope sheave.

3. A spindle is a short shaft that imparts motion either to a cutting tool (e.g. drill press spindles) or to a work piece (e.g. lathe spindles).

14.2 Material Used for Shafts

The material used for shafts should have the following properties :

1. It should have high strength.
2. It should have good machinability.
3. It should have low notch sensitivity factor.
4. It should have good heat treatment properties.
5. It should have high wear resistant properties.

The material used for ordinary shafts is carbon steel of grades 40 C 8, 45 C 8, 50 C 4 and 50 C 12. The mechanical properties of these grades of carbon steel are given in the following table.

Table 14.1. Mechanical properties of steels used for shafts.

Indian standard	Ultimate tensile strength,	Yield strength,
40 C	560 -	32
45 C	610 -	35
50 C	640 -	37
50 C	700	39

When a shaft of high strength is required, then an alloy steel such as nickel, nickel-chromium or chrome-vanadium steel is used.

14.3 Manufacturing of Shafts

Shafts are generally manufactured by hot rolling and finished to size by cold drawing or turning and grinding. The cold rolled shafts are stronger than hot rolled shafts but with higher residual stresses. The residual stresses may cause distortion of

14.4 TYPES OF SHAFT

the shaft when it is machined, especially when slots or keyways are cut. Shafts of larger diameter are usually forged and turned to size in a lathe.

The following two types of shafts are important from the subject point of view :

1. Transmission shafts. These shafts transmit power between the source and the machines absorbing power. The counter shafts, line shafts, over head shafts and all factory shafts are transmission shafts. Since these shafts carry machine parts such as pulleys, gears etc., therefore they are subjected to bending in addition to twisting.

2. Machine shafts. These shafts form an integral part of the machine itself. The crank shaft is an example of machine shaft.

14.5 Standard Sizes of Transmission Shafts

The standard sizes of transmission shafts are :

25 mm to 60 mm with 5 mm steps; 60 mm to 110 mm with 10 mm steps ; 110 mm to 140 mm with 15 mm steps ; and 140 mm to 500 mm with 20 mm steps.

The standard length of the shafts are 5 m, 6 m and 7 m.

14.6 Stresses In Shafts

The following stresses are induced in the shafts :

- 1.** Shear stresses due to the transmission of torque (i.e. due to torsional load).
- 2.** Bending stresses (tensile or compressive) due to the forces acting upon machine elements like gears, pulleys etc. as well as due to the weight of the shaft itself.
- 3.** Stresses due to combined torsional and bending loads.

14.7 Maximum Permissible Working Stresses for Transmission Shafts

According to American Society of Mechanical Engineers (ASME) code for the design of transmission shafts, the maximum permissible working stresses in tension or compression may be taken as

- (a)** 112 MPa for shafts without allowance for keyways.
- (b)** 84 MPa for shafts with allowance for keyways.

For shafts purchased under definite physical specifications, the permissible tensile stress (σ_t) may be taken as 60 per cent of the elastic limit in tension (σ_{el}), but not more than 36 per cent of the ultimate tensile strength (σ_u). In other words, the permissible tensile stress,

$$\sigma_t = 0.6 \sigma_{el} \text{ or } 0.36 \sigma_u, \text{ whichever is less.}$$

The maximum permissible shear stress may be taken as

- (a)** 56 MPa for shafts without allowance for key ways.
- (b)** 42 MPa for shafts with allowance for keyways.

For shafts purchased under definite physical specifications, the permissible shear stress (τ) may be taken as 30 per cent of the elastic limit in tension (σ_{el}) but not more than 18 per cent of the ultimate tensile strength (σ_u). In other words, the permissible shear stress,

$$\tau = 0.3 \sigma_{el} \text{ or } 0.18 \sigma_u, \text{ whichever is less.}$$

14.8 Design of Shafts

The shafts may be designed on the basis of

- 1.** Strength, and **2.** Rigidity and stiffness.

In designing shafts on the basis of strength, the following cases may be considered :

- (a)** Shafts subjected to twisting moment or torque only,
- (b)** Shafts subjected to bending moment only,

- (c) Shafts subjected to combined twisting and bending moments, and
- (d) Shafts subjected to axial loads in addition to combined torsional and bending loads.

We shall now discuss the above cases, in detail, in the following pages.

14.9 SHAFT SUBJECTED TO TWISTING MOMENT ONLY

When the shaft is subjected to a twisting moment (or torque) only, then the diameter of the shaft may be obtained by using the torsion equation. We know that

$$\frac{T}{J} = \frac{\tau}{r} \quad \dots(i)$$

where

T = Twisting moment (or torque) acting upon the shaft,

J = Polar moment of inertia of the shaft about the axis of rotation,

τ = Torsional shear stress, and

$$T = \frac{P \times 60}{2\pi n}$$

$$T = \frac{\pi}{16} \times \tau \times D^3$$

$$r = \frac{d}{2}$$

$$J = \frac{\pi D^4}{32}$$

$$\frac{32T}{\pi D^4} = \frac{2\tau}{D}$$

3.3.4 Cutting Force of Industrial Paper

The cutting force for industrial paper occurred when the blade force is larger than the tear force of industrial paper. In this industrial shredder project, there are 56 blades in each shaft and 112 cutting blades in both shafts respectively. Each blade required 0.27N force to cut industrial-size paper. The total cutting force for this industrial paper shredder is found out as:

$$F_{\text{cutting}} = 0.27 \times C \quad (3.5)$$

Where C represents the total number of blades in a paper shredder.

3.4.5 Number of Papers in Each Cut

The industrial paper cutting force is found out by using the following relation:

$$F = 2\pi \times n \times p \quad (3.6)$$

Where “n” represents the number of papers in each cut and “p” is the pressure force to cut industrial paper that was equal to 0.789 N/m².

3.4.6 Results

$N_{\text{motor}} = 6690 \text{ RPM}$

Motor Power = P = 0.588 kW = 588W

Cutting force for each blade = 0.27N

Total number of blades in industrial paper shredder = 112

Pressure force = p = 0.789 N/m²

Step (1):

We have the motor speed and the product of transmission and receiving teeth. We used equation (3.1) to find out the speed of industrial paper shredder.

$$\text{out put paper speed}(N_3) = 6690 \times \frac{(37 \times 9)}{(7 \times 28 \times 6 \times 41)}$$

$$N_3 = 40.303 \text{ RPM}$$

Speed of industrial paper shredder = speed of third gear = $N_3 = 40.303 \text{ RPM}$

Step (2):

Now, we used equation (3.4) to find torque on the shaft.

$$T = \frac{588 \times 60}{2 \times 3.14 \times 40.303}$$

$$T = 139.39 \text{ N.m}$$

Torque on the shaft = $T = 139.39 \text{ N.m}$.

Step (3):

The total cutting force for industrial paper shredder is found out by using equation (3.5)

$$F_{\text{cutting}} = 0.27 \times C$$

$$F_{\text{cutting}} = 0.27 \times 112$$

$$F_{\text{cutting}} = 30.24 \text{ N}$$

Total cutting force for industrial paper shredder = 30.24 N

Step (4):

In order to cut industrial paper, we need to fulfill the following condition as ($F_{\text{paper}} < F_{\text{cutting total}}$).

$$F_{\text{paper}} < 30.24$$

$$\text{Maximum number of industrial paper} = F_{\text{paper}} = 6 \times 2 \times 3.1415 \times 0.789$$

$$F_{\text{paper}} = 29.729 < 30.24 \text{ N}$$

The paper shredder can be shred up to six industrial papers in each cut.

3.4.7 Summary of the Calculations

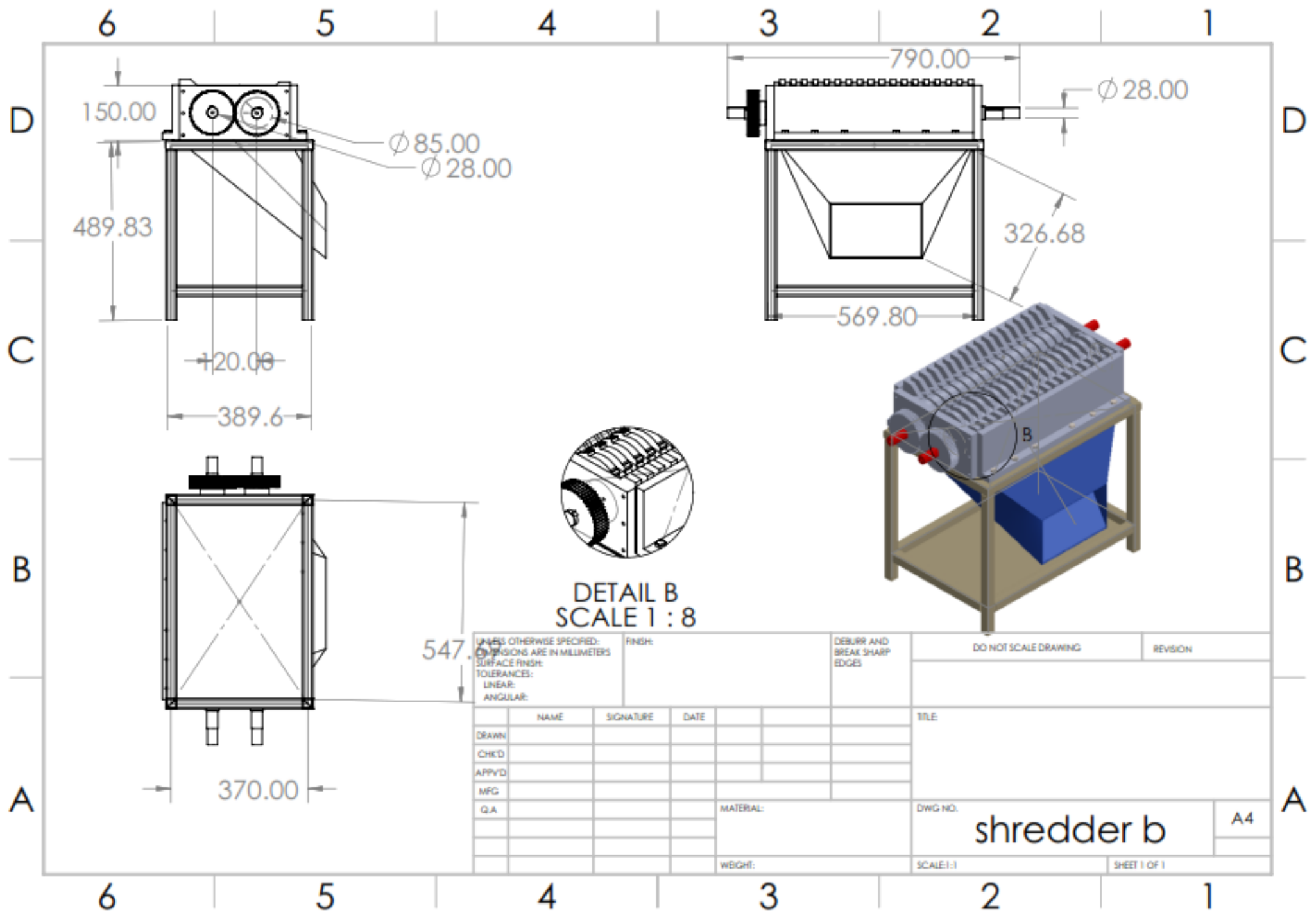
We repeated the same procedure steps of the calculations when the shredder force is equal to 0.789 N/m².

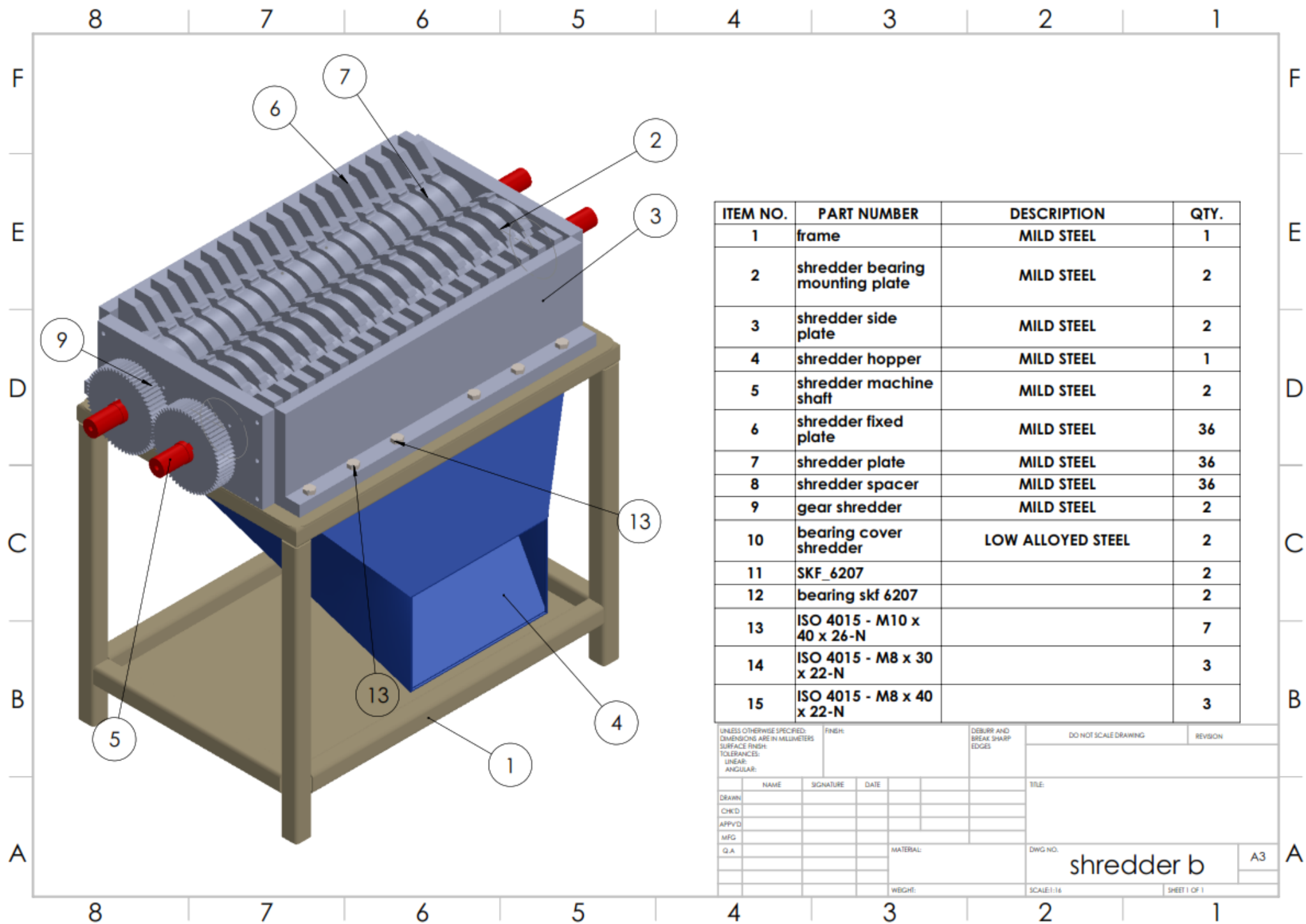
Table 3. SUMMARY OF RESULT

Shredder Features	Results
Industrial paper speed (N ₃)	40.303 RPM
Torque on the shaft (T)	139.39 Nm
The total cutting force of the shredder	30.24 N
Single cutting force	0.27 N
Number of industrial paper in each cut	6 industrial papers

3.4 Product Subsystems and selection of Components

The figure shows the exploded view of the industrial shredder project. The various parts of the shredder project including cutting blades, gears, two bases, motor, screws, and shaft. Each blade has the capacity to cut six industrial papers during each cut. Each blade contains its inner cavity that allows the shaft to slide through cutting blades. The blade also contains small teeth in its inner race because to prevent rotation without intension. Both shaft and cutting blade is made with standard AISI 304 steel. This material is a very popular choice because of its availability and low cost as well. The two bases are designed to hold gears, a motor, and two shafts. In this industrial shredder project, the gears are used for speed reduction application and transfer power from one shaft to another shaft.





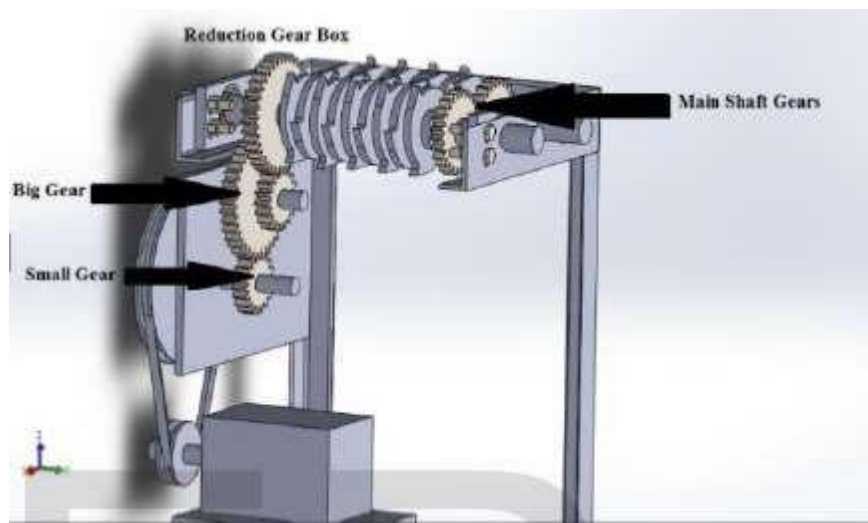


Figure 3.transmission system

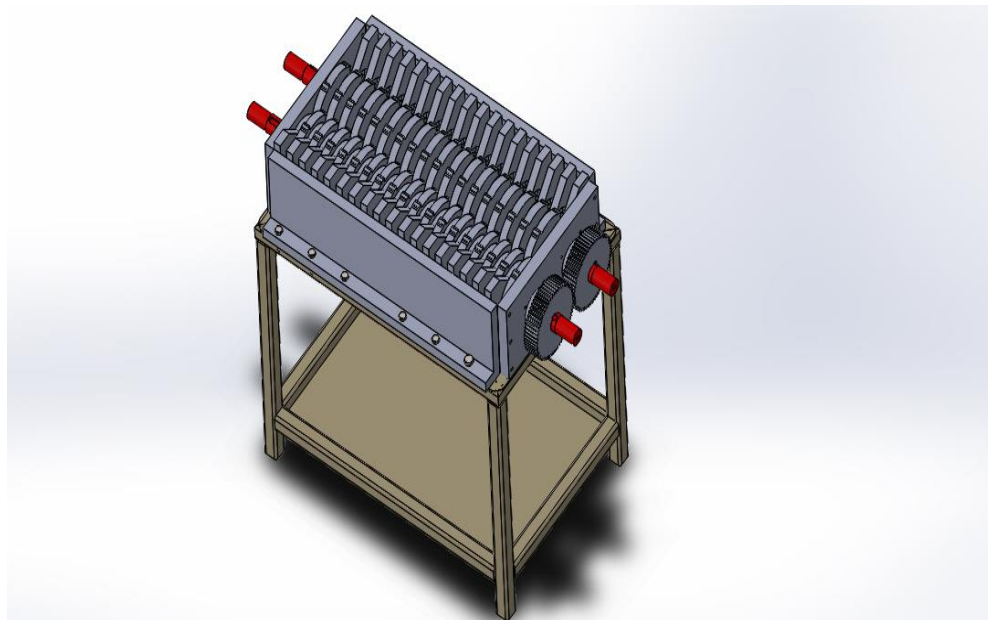


Figure 4.isometric view of shredder machine

3.5 Manufacturing and Assembly

Both shaft and cutting blade is made with standard AISI 304 steel. This material is a very popular choice because of its availability and low cost as well. The motor gear and first gear are made with plastic to minimize vibration and noise. The outer casing of the paper shredder machine was made with aluminum because it is a reliable and lightweight widely used material. This industrial paper shredder process starts with a 0.588 kW DC electric motor, which produces rotary motion through motor gear. In this paper shredder machine, the compound gear system was chosen over the pulley system due to more efficiency. The compound gear train was provided various gear ratios and less speed which is what is needed here. The different pairs of shafts are used to enhance cutting force and security as well as provide a smaller cut. The main shredder shaft is held cutting blades, which are responsible for the shredding process. For the shredding process, the cutting blades are made with stronger materials like stainless steel.



Figure 5.view point of a shredder



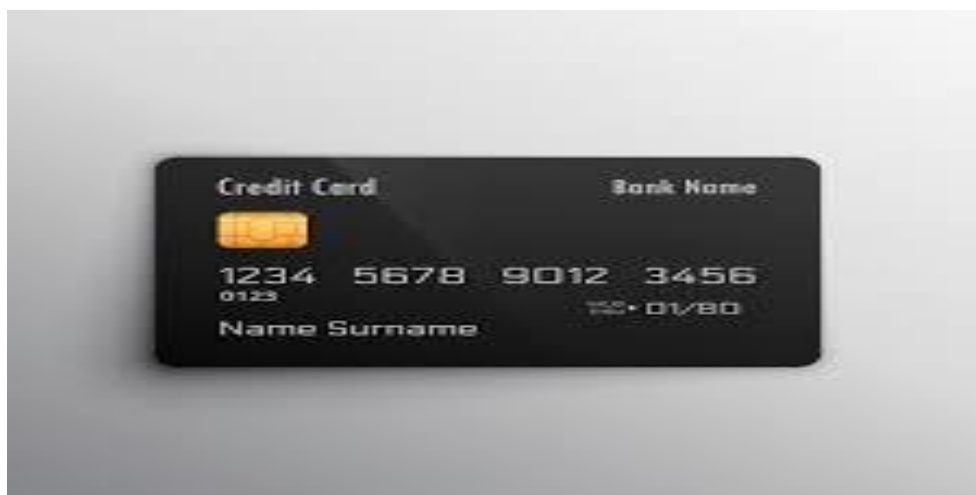
Figure 6.assembly

3.5.1) PRESENTATION OF PAPER SIZE SHEETS SHREDDED IN THE MACHINE

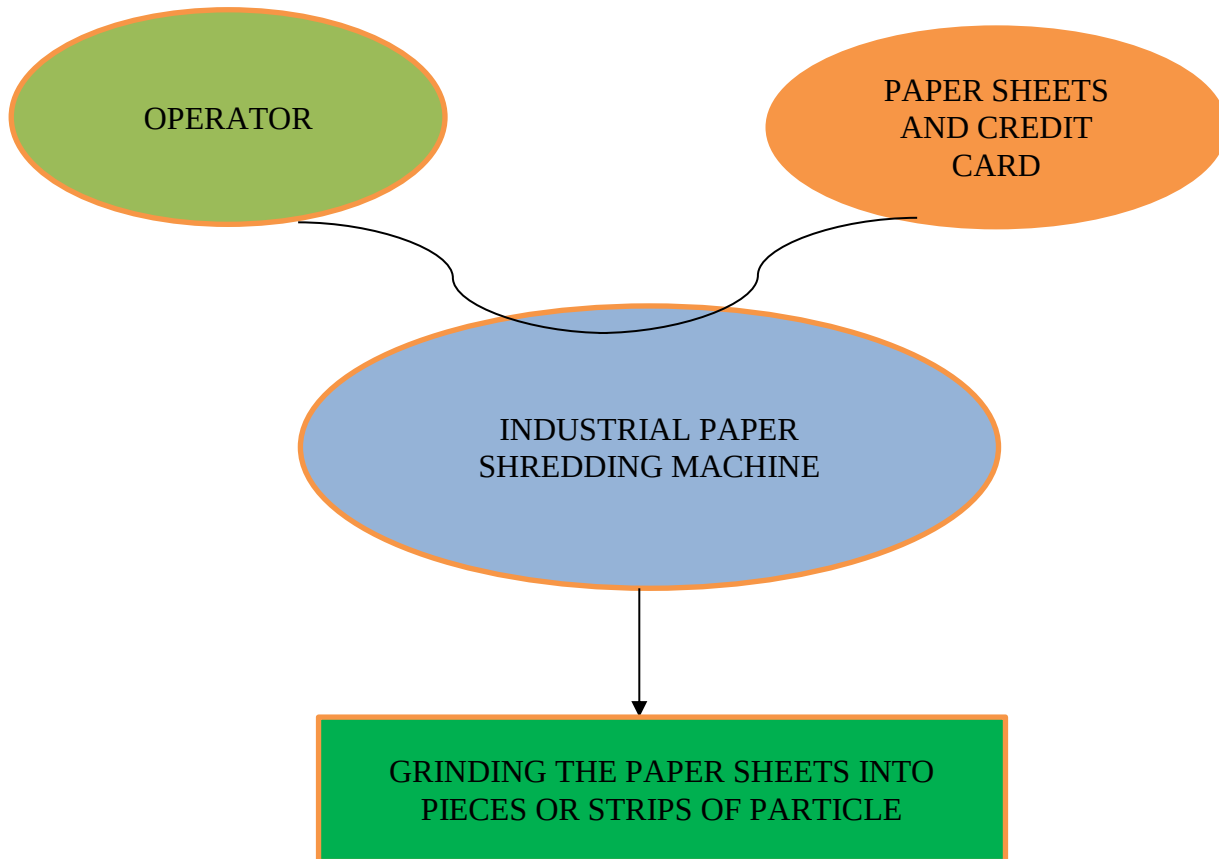
There is no standard size of paper shredded in a shredder machine, other than those set forth for governmental classified documents there is no real standard. Government approved high security shredders created for destroying top secret and classified documents, render an 8.5" x 11" sheets to 15000 particles. While impressive, this kind of destruction is rarely needed for business use especially in the case of mobile shredding. These shredding machine can shred standard paper size from A0-A10 size, credit card and other more.

PAPER SIZES A0 TO A10

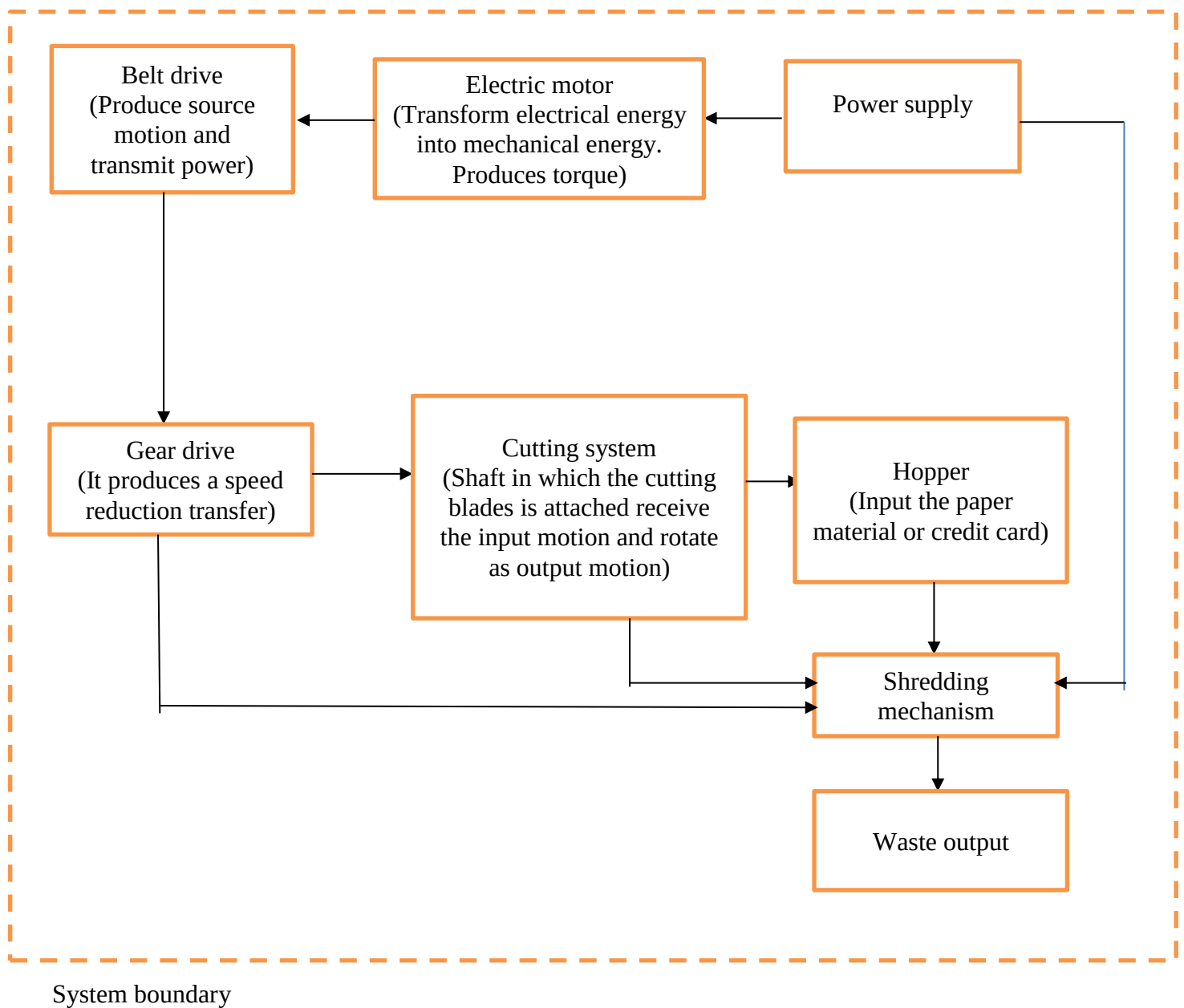
Paper	Height x Width (mm)	
A0	1189mm	841mm
A1	841mm	594mm
A2	594mm	420mm
A3	420mm	297mm
A4	297mm	210mm
A5	210mm	148mm
A6	148mm	105mm
A7	105mm	74mm
A8	74mm	52mm
A9	52mm	37mm
A10	37mm	26mm



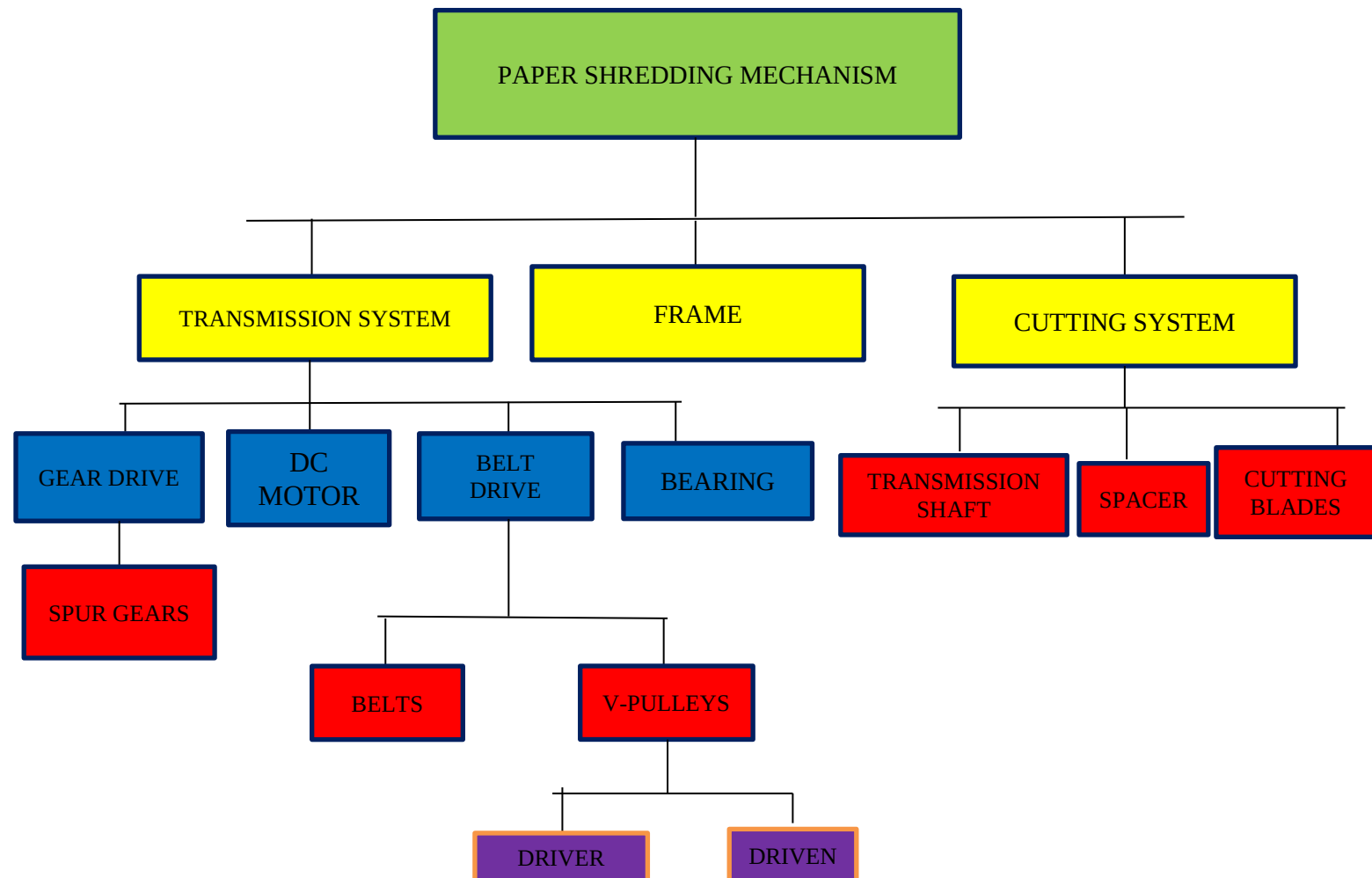
3.5.2) FUNCTIONAL SERVICE OF A PAPER SHREDDING MACHINE



3.5.3) FUNCTIONAL ANALYSIS OF A PAPER SHREDDER



3.5.4) SYSTEM STRUCTURAL ANALYSIS



- A) TRANSMISSION SYSSYTEM:** It is a system or mechanism which interpose between the source of power and specific application for the purpose of adapting one to the other. Most mechanical transmissions function as rotatory speed changers; the ratio of the out put speed to the input speed may be constant.
- B) CUTTING SYSTEM:** These system is used for shredding or cutting paper material into strips or particles

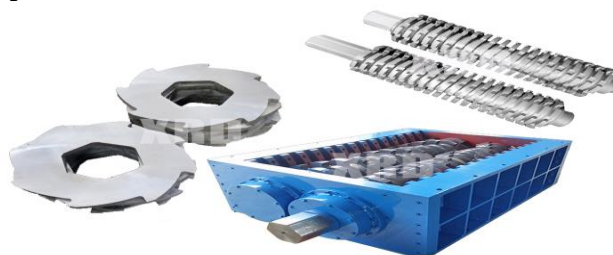


Figure 7.cutting system

- C) BELT DRIVE:** It is a frictional drive that transmits power between two or more shafts using pulleys and elastic belt. It can operate in wide range of speed and power requirements. It is also highly efficient.



Figure 8.pulley drive

- D) GEAR DRIVE :** Also know as a gear box, there 03 main functions are; to increase torque from the driving equipment (motor) to the driven equipment, to reduce the speed generated by the motor, and/or to change the direction of the rotating shafts.



Figure 9.gear drive

- E) BEARING:** It is used in the dc machine to reduce the frictional losses. Thus , the main function in the dc machine is to support the machine shaft with minimum friction. It also allow a smoother rotation.



Figure 10.bearing

F) DC MOTOR: This shredder machine uses a forward reverse motor, and the forward reverse motor often causes large sparks when the motor is working due to double steering, resulting in short life and poor EMC.

G) GEARS: There have 04 main function which are;

- Speed: different sizes gears to be used to efficiently change the speed of a system.
- Force: In similar setup, you can use gears to increase the force of a system; that extra speed doesn't appear out of nowhere
- Rotation direction
- Movement

H) TRANSMISSION SHAFT: It is a rotating machine element which either receive power, transmits power or both. Shaft are of circular cross section.



Figure 11.shaft

I) PULLEYS: It is used singly or in combination to transmit energy and motion. Pulleys with grooved rim are called v grooved pulleys.



Figure 12.pulley

3.5.5) SIMULATION ANALYSIS

Design calculation of shredder details:

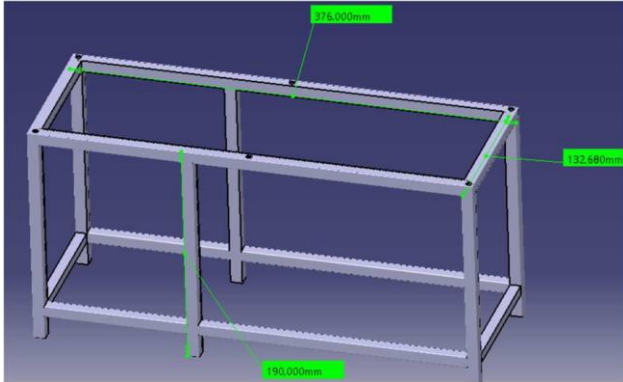


Figure 2 The tripod

Dimensions (Figure 2): 376x132.68x190 (mm) The weight of the shredder is 12 kg.
The vibration force acting on the tripod is: $P = 120 \text{ N}$.

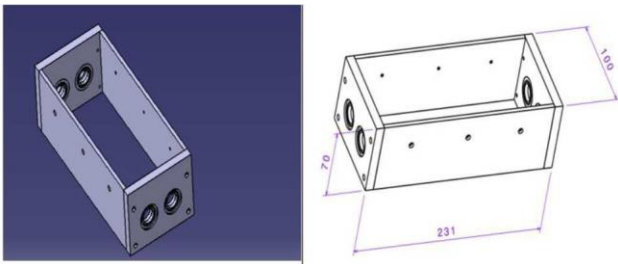


Figure 3 The frame

Frame: Supporting parts for shafts, blades and gears, fastened with bolts.

The frame (Figure 3) has dimensions of 231x100x70 mm, with length(l) = 231 mm, width(w) = 100mm and height(h) = 70 mm

2.1 Blade design

The cutting system consists of a shaft, cutting blade, washer and gear. The blade is a blade with 3 cutting edges, circular hole design with a diameter of 18 mm, mounted on the main shaft and the spindle moves together.

optimized cutting blade design concentrates tension, mass thereby reducing the load on the shaft.

Figure 3 shows the original unmodified 3D design of the blade. The cutting blade is a circular blade with three cutting edges with a round hole in the middle that attached to the spindle and moves the spindle. This blade is used to cut paper in the vertical direction. However, there was a problem of stress concentration at the bend.

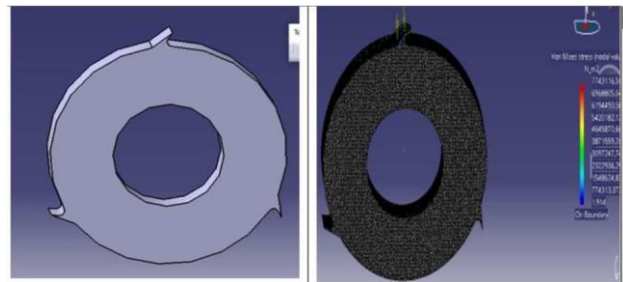


Figure 4 Original blade with stress distribution

Figure 4 shows the modifications in the blade design. Therefore, some modifications have been made, stress concentration problem was solved by creating a tangent in a side edge. However, although stress reduction has been fixed to avoid blade breakage, the mass is heavier than the blade at the first time $m = 0.049 \text{ kg}$ (Figure 5).

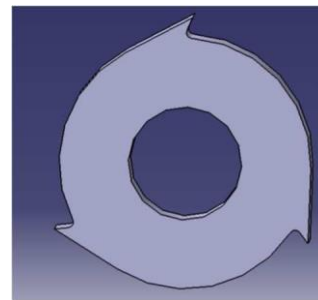


Figure 5 Blade redesigned the first time

Figure 6 shows in this revision, the weight of blade has been reduced. In the first, two blades are made with a circle of diameter $d = 40 \text{ mm}$ and with a single center. In the design the size has been reduced to $m = 0.047 \text{ kg}$, instead of a radius of 20 mm with a common center, this blade is designed in such a way that it creates arcs of 20 mm radius and are three different arcs. whose center is the vertex of an equilateral triangle, is the center of the hole attached to the major axis. Blade thickness equal to 6 mm, made of steel and

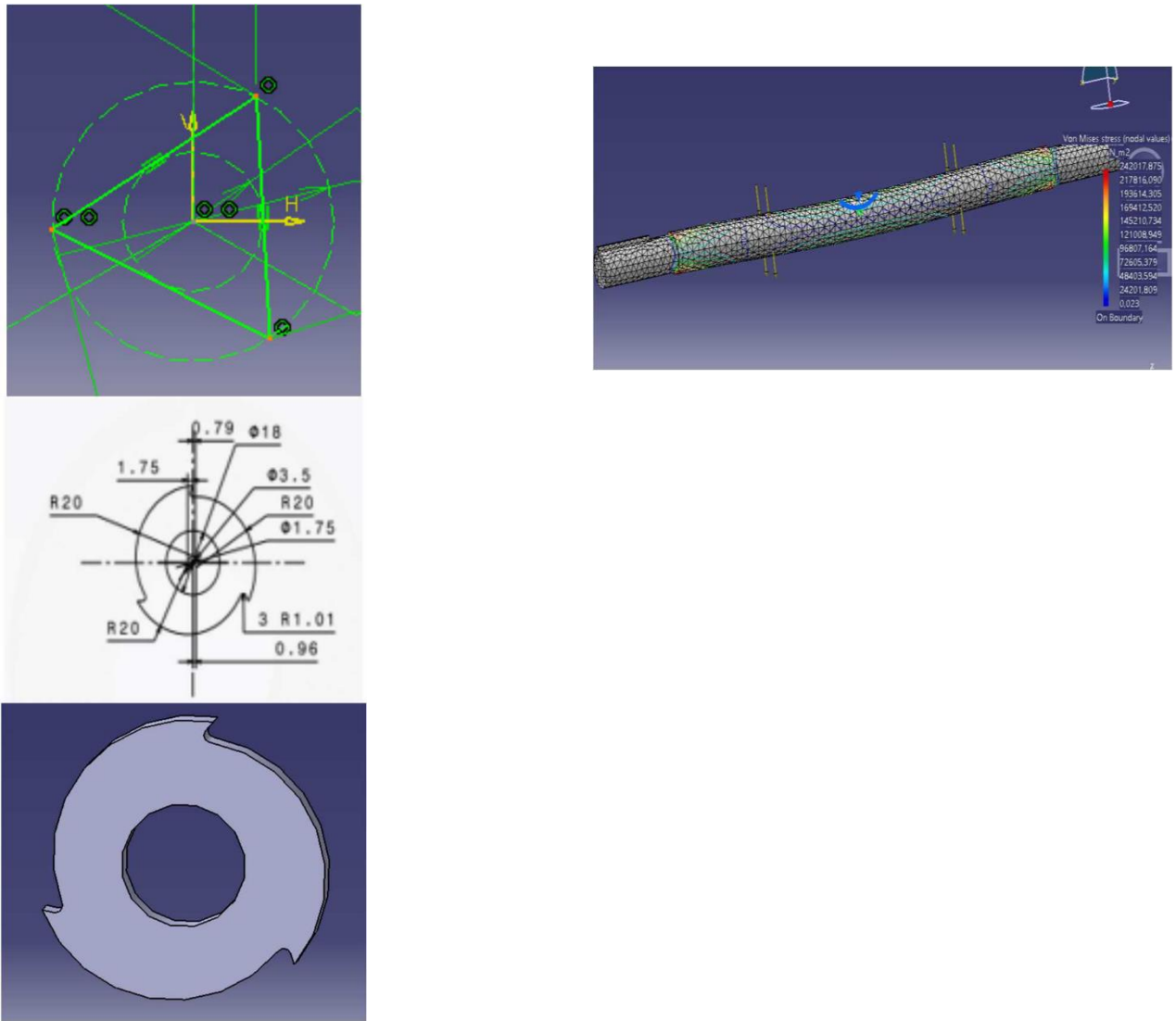


Figure 6 Final design shape cutting blade

The power calculation

The engine speed can be determined from the shear stress of the paper and the amount of paper. Shear stress is the result when a force is applied to a product that cause shear deformation of the material in a plane parallel to the direction of force application. When cutting paper, the shear stress value of the blade is greater than the allowable shear stress value of the paper

Shear stress is express.

3.5.6) MATERIALS USED FOR THERE PAPER SHREDDING MACHINE

S/N	Machine component	Material selected	Selection criteria
1	Frame	Malleable steel	Adequate strength, hardness, toughness, readily available
2	Belt	Rubber	Good corrosion resistance, low cost and readily available
3	Hopper	Galvanised mild steel plate	Adequate Strength, corrosion resistance, cheap and readily available
4	Auger screw shaft	Galvanised mild steel	Adequate strength, cheap and readily available
5	Shredder shaft	Mild steel	Adequate strength, cheap and readily available
6	Shredder shaft blade	High speed steel	High strength, toughness, wear resistance and readily available
7	Die plate and breaker plate	Stainless steel	Adequate strength, corrosion resistance, and readily available
8	Extrusion barrel	Galvanised mild steel	Adequate Strength, corrosion resistance, cheap and readily available
9	Pulley	Mild steel	Adequate strength, cheap and readily available
10	Motor cover	Mild steel	Adequate strength, cheap and readily available

2.4. Design Analysis

Figure 13.material list

where, ω_{ss} is the angular velocity of the shredder shaf

3.6 3D CAD DESIGN

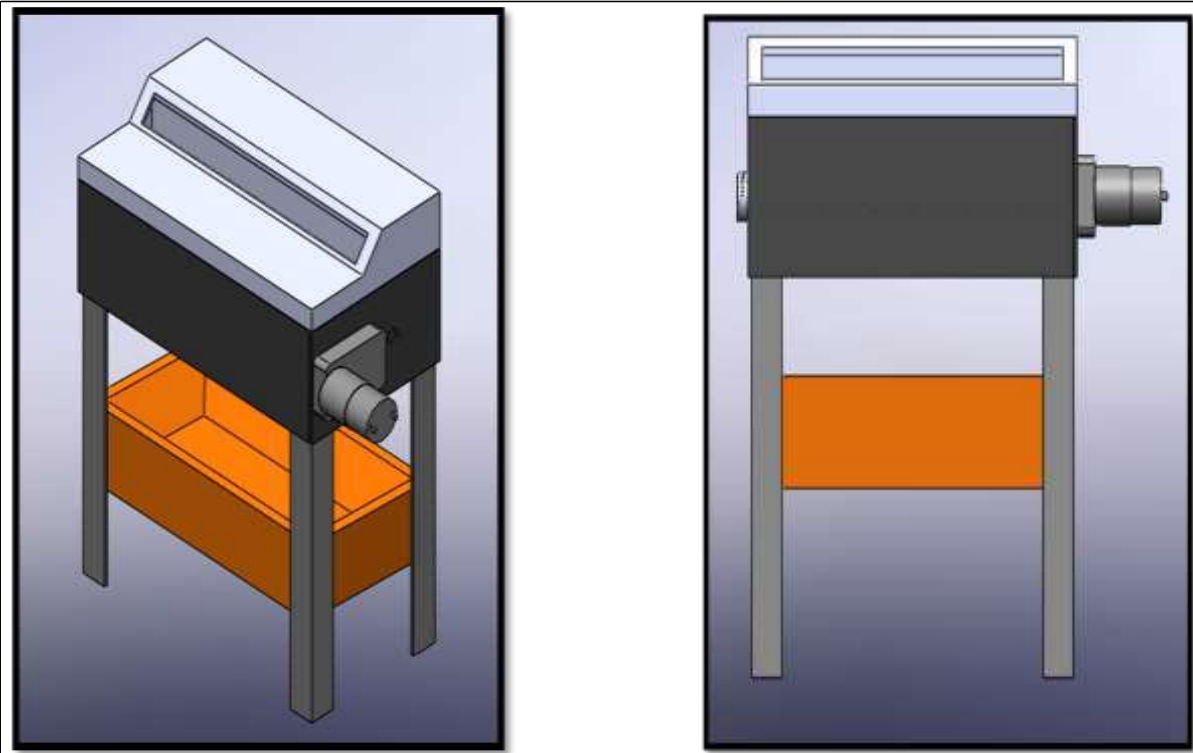


Figure 14.rigth side view & front view

3.7 Economic Evaluation of projects

The economic evaluation of a project is considered the most important phase of any report because it explains the viability of a project. When a company plans to launch a news service or a new product in the market the most important thing which is considered by the deciding body of that organization is a financial analysis of a project which includes the 10-year or more than 10-year plan about the working procedures and profits expected from that product. This financial analysis helps the company to include both the equipment and construction cost as a part of the capital cost that will be required as a non-recurring cost needed to be paid by the investors of the organization. The operating cost included in the economic evaluation has

consisted of energy cost, electricity cost, labor cost, insurance cost, and the cost of taxes. All these operating costs are the recurring costs that need to be paid continuously by the company till the end of the total tenure linked with that project.

After this, the revenue generated by the company each year with the help of different products is incorporated. The revenue section includes the total income that is generated by the company with the help of its products. If the company has incorporated more than one product then the sum of incomes generated by all those products will be equal to the revenue generated by the company in that particular year. After the calculation of revenue, the next end most important procedure for the economic evaluation of any project or product is to calculate the profit generated by that company with the help of the aforementioned products as it can be one product or several products. The profit generated by the company is calculated by subtracting the annual cost, both capital and operating cost, from the revenue generated by that company. The economic evaluation for our product is performed and shown in the chart below. the costs involved in the project are used with the following values:

Capital cost

equipment cost = 340,000frs

construction cost = 68,000frs

Operating cost

energy cost = 136,000frs

electricity cost = 81,600frs

The annual revenue is expected to be equal to 816,000FRS.

The complete chart incorporating the costs, revenue generated, and the profits are shown below:

Table 4.COST ESTIMATION

	year 1	year 2	year 4	year 5	year 6	year 7	year 8	year 9	year 10	
capital cost(FRS)										
Equipment	340000	0	0	0	170,000	0	0	0	0	
Construction	68000	0	0	0	34000	0	0	0	0	
Total	408000	0	0	0	204000	0	0	0	0	
Operating cost(frs)										
Energy	0	136000	136000	136000	136000	136000	136000	136000	136000	
Electricity	0	81600	81600	81600	81600	81600	81600	81600	81600	
labour	0	0	0	0	0	0	0	0	0	
Insurance	0	0	0	0	0	0	0	0	0	
Taxes	0	0	0	0	0	0	0	0	0	
TOTAL	0	217600	217600	217600	217600	217600	217600	217600	217600	
Revenue(frs)										
Product 1	0	816000	816000	816000	816000	816000	816000	816000	816000	
Product 2										
Product 3										
TOTAL	408000	816000	816000	816000	816000	816000	816000	816000	816000	
PROFIT(FRS)	-408000	598400	598400	598400	394400	598400	598400	598400	598400	

The operating cost for 0th year is not included because the project started it is working from the start of year one after the completion of its construction and design. Therefore, the 0th year incorporates only the capital cost which is a nonrecurring one. After that, capital cost will not be included in the further years because the product is designed and completed only once at the start of a project. in the 6th year, the capital cost is once again mentioned but quite less as compared to the first one is showing the maintenance cost that is required to be paid by the investors for effective working and efficient process flow of the product. Because in our case, we have only one product that needs to be considered, therefore the revenue generated by that product is mentioned. Then, the values for profit are calculated by subtracting the costs paid by the company and the revenues generated in response to that.

The cash flow generated in this regard with the help of applying the values calculated for our product, the shredder machine, is shown below:

Chapter 4: System Testing & analysis

4.1 Sounds and vibration test results [9][10]

The analysis for the sound and vibration on the as-built model is as follows

Table 5.SOUND & VIBRATION RESULT

Part	Natural Frequency (Hz)	Test Frequency (Hz)
Shaft	Approx. 315	80
Electric Motor	Approx. 260	65
Cutting blades	Approx. 246	62

The above vibration analysis results indicate that the machine is and will operate within the safe region and might not break down before its service time.

4.2 Decision matrix [10][11]

The design criteria are being influenced by the following decision matrix. The matrix assigns the weightage of the criteria and appropriate score as far as the present project is concerned. The sum of all weights is 100% and the Score is from 1-10 where 1 is the least and 10 is the most.

Criteria	Weightage (%)	Score (1-10)
----------	---------------	--------------

Table 6.DECISION MATIX ANALYSIS

Reliability	15	8
Cost effective	10	6
Ease of operation	10	9
Speed	15	6
Stability	20	8.5
Portability	10	5
Noise	15	8.5
Size of collecting bin	5	9

4.3 Cost analysis

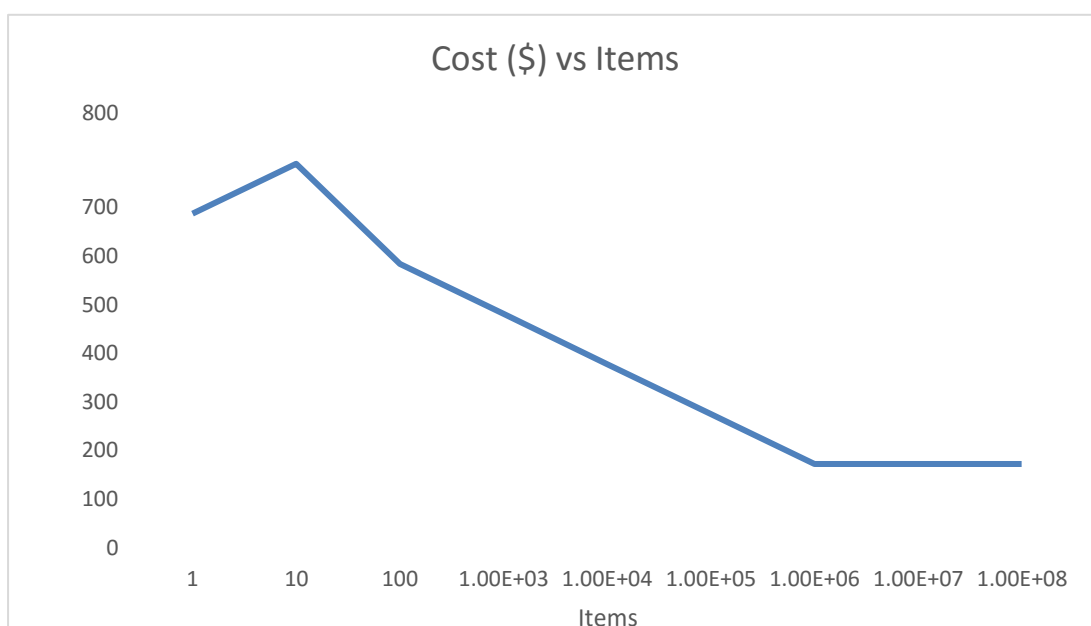
The following table gives the cost of different parts and the associated graph

Table 7.COST ANALYSIS

Part / Service	Cost (frs)
Cover Sheet & stand	57800
Blades	170,000
Shaft(s)	34000
Gears	51000
Motor	30,000
Assembly work	68000



From the above graph, it could be observed that the blades have a lion-share of the total costs followed by the assembly work. But overall work is being done cheaply (around 600 USD). Besides, the current project is only a proof of concept and the costs might come down as economies of scale assume the position with large-scale industrial manufacturing. It is assumed that the cost of the final product with scaling of the economy could be seen in the following diagram.



Chapter 5: Project Management

5.1 Project planning & Monitoring

Project planning in the context of the present project is done with the help of the project planning module of MS Excel. The project scope is defined in the beginning. Thereafter the tasks are distributed among team members taking into account their skills and interests. The project management plan could be seen from the following Gantt chart. It is from this Gantt chart project deliverables are monitored closely.

5.2 members contribution

The team members met every other day to discuss the progress and brainstorm ideas for efficient and effective project management. The discussions could range from theoretical analysis to economic considerations. The contribution of each member could be found from the Gantt chart in the above section and also as a brief table below. Note that the numbers are subjective and by no means the perfect representation of member's contributions.

5.3 Project Challenges

5.3.1 Design challenges overview

One of the biggest challenges the project has is product specification standards. Although a lot of standards exist for shredded paper, no universal standard exists for components for a paper shredder. In addition to that, papers come with different standards in which they are eligible to be recycled. Also, the same level of shredded paper could be accomplished with different blade designs, materials, and types. Choosing a proper design based on the constraints is a huge task.

5.3.2 Fabrication challenges overview

The obvious design challenge is communication in the present pandemic i.e., COVID-19. Although the internet helps a lot to have online video meetings, some situations require one-on meetings. These are accomplished by adhering to proper pandemic protocols.

The fabrication of the as-built model requires a safe place to assemble, place workshop tools and fabricate using a variety of techniques such as pressing, welding, drilling, etc. Added to those safety requirements are the restrictions imposed due to the pandemic COVID-19. Team members chose one vaccinated person to do all the work related to purchases and rentals while other members maintained the proper safety protocols. Online purchases are made whenever possible.

Chapter 6: Project Analysis

6.1 Long term learning

The current project is a combination of theory and practice in that it asks for

1. Design
2. Analysis
3. Fabricate
4. Test

All with a single goal of application i.e., ‘Paper shredder’.

The concepts learned had vast implications on the team member's understanding of the vast difference between the proper understanding of a theory and its practical implementation. E.g., team members understood proper time and work management during this project, while enhancing their technical and people skills. This learning goes a long way in understanding complex projects in the future.

6.2 Impact of Engineering Solutions

The solutions proposed in this project have a small yet focused impact on the team and society at large as it shows how the engineering could be done with the help of CAD and practical testing and analysis. As far as the impact of the project is considered, proper ‘Paper Shredder’ which could shred paper and soft materials alike have a huge potential. As explained elsewhere in the project, paper although a thing of the past is widely used to control knowledge dissemination. It becomes of obvious importance that the papers which carry official secrets

and/or of secret in nature should be shredded without a chance of reconstruction. Similarly, soft paper-like materials, e.g., credit cards, etc. also needed to be destroyed while maintaining the feasibility to be recycled.

6.3 Contemporary Issues Addressed

The Kingdom of Saudi Arabia similar to other emerging economies utilizes paper-based information transmission. The recycling of paper requires a lot of energy if the paper is not shredded. Additionally, not all information is needed to be known to everyone. By proposing a neat solution like a custom paper shredder, employees carrying information could be encouraged to destroy the secrets as soon as their purpose is accomplished. It also helps to inculcate a feeling of cleanliness in employees and encourage them to keep their environment clean.

Chapter 7: Conclusions and Future Recommendations

7.1 Conclusions

The current project is one small step in the right direction of conservation of the environment and preservation of important information. The project started with well-defined objectives and then specifications are estimated using theoretical knowledge and expertise. A brief literature review comparing previous work in the same direction is being presented. A comprehensive design involving theoretical calculations and CAD design is being performed. The fabrication of 'Paper Shredder' is followed by basic testing and analysis. An overview of project management and project analysis finishes this project.

7.2 Future Recommendations

There is a scope to improve the current project by integrating dynamic analysis and Computer-Aided Manufacturing. Also, the material for the project could be improved by proper research in metallurgy and material sciences. Finally, the fabrication could have been done with Computer-Aided Manufacturing to maintain proper tolerances and dimensions.

7.3) MAINTENANCE PLAN FOR THE SHREDDER MACHINE

7.3.1) PREVENTIVE MAINTENANCE

These are the action or procedure carryout on a machine or system to avoid failure or malfunction before it downtime period. These type of maintenance is labour intensive and require a lot of planning. These planning include routine checkup carryout daily, monthly and yearly. There following routine for preventive maintenance are :

7.3.2) DAILY CHECKS (OPERATOR)

- Test the shredder safety equipment(emergency stop buttons, etc) prior to use and only use if all the safety features are functioning correctly.
- After the final run of the day, clean the shredding surfaces.
- Lubricate all the moving part with light machine oil and check for smooth operation.
- Ensure all safety barriers are in place and secure to effectively reduce risk from moving parts.
- Before the shredder is used, ensure correct positioning of the paper under shredder to capture material.
- Report all faults, record these in the user log; and assign each fault with a unique, sequential number.
- Proper and regular cleaning is essential, since debris may cause the shredder to be malfunction.
- Never attempt to clear a blockage while machine is switched on or connected to the power supply.

7.3.3) weekly maintenance (operator)

Check that the following are working:

- Indicator lights
- Seal and locks
- Inspect, lubricate and tighten all joints
- Ensure that all safety barriers are in place
- Report all faults; record them in the operation log; and assign each fault with unique, sequential number. These check must be recorded by operator or responsible person in the shredder maintenance log. Any failures that cannot be rectify by operator must be reported to the department manager or other designated responsible person. Unsafe or faulty equipment must not be used and where there are multiple users, appropriate signage and means to lock out should be employed to prevent its use.

7.3.4) MONTHLY INSPECTIONS AND QUARTERLY MAINTENANCE

The following checks and activities should be undertaken by a maintenance or service engineer:

- ❖ Check the user and maintenance log for signs of problem requiring action.
- ❖ Inspect air filters if any where fitted and replace according to schedule and label the new filter with the date of installation and/or date of replacement.
- ❖ Inspect the blades surfaces for wear and sharpen or replace when necessary.
- ❖ Check oil level, top up and replace according to the manufacturer's requirement.
- ❖ Ensure that all safety barriers are in place, are secure and effectively protected from accessing moving parts during operation.

These inspections, any failure and any subsequent maintenance must be recorded in the maintenance log and signed off by the personnel conducting the work.

7.3.5) ANNUAL MAINTENANCE AND INSPECTION

The following checks and activities should be undertaken every 12 months by a maintenance or service engineer:

- ✓ Check that all recommendations made at the previous inspection have been correctly carried out.
- ✓ Check service history for recurring faults and corrective action to determine if repairs, refurbishment or replacement is required.
- ✓ Check conditions of the seals and shredding surfaces, repair or replace where necessary.
- ✓ Test all function under working conditions to the satisfaction of the operator or responsible person.
- ✓ Ensure that all safety barriers are in place, are secure and effectively reduce risk from moving parts.

These inspection, any failure and any subsequent maintenance must be recorded in the maintenance log and signed off by the personnel conducting the work.

7.4) BREAK DOWN REPAIR OF SHREDDER

If the paper cannot be shred or the shredder makes strange noise , do you notice an unusual vibration coming , or is the machine refusing to start , these are the cause and solutions measure response:

- Your shredder is making strange sounds; do you hear a grinding or knocking sound you don't know the cause, but it ensure doesn't seem normal. Most likely, small pieces of metal or stones are caught in the rotor or counter knives of the machine. To find out, stop the shredder and switch it to service mode. This allows you to unlock and open the maintenance doors. The procedures depends on the type of your shredder machine. For an exact description, check the operating instructions. Here it is very important to examine the condition of the shredding tools and counter blades. They might be damaged by foreign objects, in which case replacement will be necessary.
- Your machine is reversing too often; what does it mean when machine's main drive frequently goes into reverse, turning first in one direction and then in the other, This is not necessarily a reason to call the service hotline. As a rule, it means that shredding tools and counter blades are worn. Simply rotate the tools or replace them if necessary. However, depending on the model series, the rotor or cutting frame may be also be showing signs or wear. Then you'll have to check them more closely and take further action.
- Bearing are very hot; does your control panel show temperatures exceeding 80° celcius in your shredder's bearing, these have several causes:
 1. Lubrification is insufficient. Check whether oil or grease has leaked. Replace the seals and apply new grease.
 2. Dirts has collected in the bearings. This causes increased friction and higher temperatures. Clean the bearing and re-grease them. Tip: Always check the

bearing for errors in assembly. This too can be a reason for the high temperatures.

- The rotor is blocked; was your shredder running smoothly until the rotor suddenly came to hard stop, causing the machine to go into fault mode, switch it to service mode and open the maintenance door. Then follow the instructions in the operating manual. If necessary, empty the shredding chamber and remove any large pieces of metal or stones. Check the shredding tools and counter blades for damaged. If the rotor in your granulator is blocked, the cause may be different than with a shredder:
 1. The vee-belt might be slipping. In this case, retighten or replace it.
 2. Material is being fed too fast to the machine. Reduce the speed.
 3. The plastic parts or paper fed in might be too broad. Only feed material of a suitable size.
- The particle size of the output material is too large; What should you do if the output of the shredder is unsatisfactory because the final particles are too large, no problem because these are some simple solutions. With a shredder there are two possible causes:
 1. The shredding tool is worn, increasing the distance to the counter blades. In the case, rotate the tool or replace it. And don't forget to correct the cutting gap if necessary. Are the final particles from your shredder too large? The cause might be blunt cutting bars, in which case you should resharpen or replace them. However, it might also be that blade gap is too large. Set it to the correct distance. The particles size can also be negatively affected if the temperature in the grinding chamber is too high. In this case, reduce the material feed.

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